

[illegible]

WS2812 Driver

The diagram illustrates a WS2812 Driver circuit. It features a +3.3V supply connected to the WS2812_DO pin via a 141 ohm resistor (R23). The signal line from the DO pin passes through a BSS138 MOSFET (Q3) and a 100 ohm resistor (R25) to the WS2812 module. The MOSFET's gate is connected to a +5V supply through a 141 ohm resistor (R24). The MOSFET's drain is connected to the WS2812 module's VCC pin through a 100 ohm resistor (R25). The MOSFET's source is connected to ground through a Zener diode (D7) and a 100 ohm resistor (R25). The WS2812 module has two 5-pin headers: P2 and P3. P2 is connected to a +5V supply through a 100 ohm resistor (R25). P3 is connected to ground through a 100 ohm resistor (R25). The WS2812 module is labeled 'WS2812 Cn.' and 'WS2812 Cn.'.

Ethernet PHY

The schematic diagram illustrates the wiring for an Ethernet PHY (ENC28J60-1/ML) IC. The IC is connected to a +3.3V power supply and GND. Key connections include:

- Power and Ground:** VDD (24) and VDDPLL (16) are connected to +3.3V. VSS (26) and VSSPLL (17) are connected to GND.
- Reset:** RESET (6) is connected to +3.3V through a 1k resistor (R3).
- Crystal Oscillator:** A 25MHz crystal (Y1) is connected between OSC1 (19) and OSC2 (20). OSC1 is also connected to VDDOSC (21). OSC2 is connected to GND. 16pF capacitors (C2, C3) are connected from OSC1 and OSC2 to GND.
- Control Signals:** SI (3), SO (2), SCK (4), CS (5), INT (28), and WOL (1) are connected to their respective external signals (ENC_SI, ENC_SO, ENC_SCK, ENC_CS, ENC_INT, ENC_WOL).
- LEDs:** LEDA (23) and LEDB (22) are connected to LEDA_CATH and LEDB_CATH.
- Transceiver Pins:** VDDRX (15), TPIN+ (9), TPIN- (8), VSSRX (7), VDDTX (11), TPOUT+ (13), TPOUT- (12), and VSSTX (14) are connected to +3.3V or GND.
- Resistors and Capacitors:** RBIAS (10) is connected to +3.3V through a 2.32k resistor (R8). VCAP (25) is connected to +3.3V through a 10uF capacitor (C11). A series of four 0.1uF capacitors (C7, C8, C9, C10) are connected in parallel between the +3.3V supply and GND.

Raspberry Pi Header

sheet56B34F8B

5V_SUPPLY 5V_MCU

File: ../../../../tmp/git/PoEPi-hardware/ws2812-controller/IdealDiode.sch

P1

3 GPIO02/I2C1_SDA VPP_(5v) 2

5 GPIO03/I2C1_SCL VPP_(5v) 4

8 GPIO14/UART_TXD VDD_(3.3v) 1 X

10 GPIO15/UART_RXD VDD_(3.3v) 17 X

ENC_SI 19 GPIO10/SPL_M0SI GND 6

ENC_SO 21 GPIO09/SPL_MISO GND 9

ENC_SCK 23 GPIO11/SPL_SCLK GND 14

ENC_CS 24 GPIO08/SPL_CE0 GND 20

26 GPIO07/SPL_CE1 GND 25

WS2812_DO 7 GPIO04/GPCLK0 GND 30

29 GPIO05 GND 34

31 GPIO06 GND 39

32 GPIO12

33 GPIO13

36 GPIO16

11 GPIO17 ID_SC 28

12 GPIO18 ID_SD 27

35 GPIO19

37 GPIO20

40 GPIO21

15 GPIO22

16 GPIO23

18 GPIO24

ENC_INT 22 GPIO25

37 GPIO26

13 GPIO27

Raspberry_Pi+_Conn

[illegible]

Linear Regulator

The schematic diagram illustrates a linear regulator circuit. The input voltage is +5V, connected to the VIN pin (pin 3) of the LM1117IMPX-ADJ (U3). The output voltage is +3.3V, connected to the VOUT pin (pin 2) of U3. The ADJ/GND pin (pin 1) is connected to ground through a 300 ohm resistor (R13). The output of the regulator is connected to a 240 ohm resistor (R15) and a 0.1uF capacitor (C19) in parallel, which is then connected to a 1uF capacitor (C20) in parallel with the output. The output is also connected to a 360 ohm resistor (R18) and an LED (D6) through a 360 ohm resistor (R18).

License: CERN OHL V1.2		
Author: Julien Marcus		
OpenFet		
Sheet: /		
File: working.kicad_sch		
Title: Raspberry Pi – Power Over Ethernet & PHY		
Size: A3	Date: 2016-02-02	Rev: 0.1d
KiCad E.D.A. eeschema 7.0.6		Id: 1/2

1	2	3	4	5	6
A					A
B					B
C					C
D					D
1	2	3	4	5	6

Sheet: /sheet56B34F8B/ File: Ideal_Diode.sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. eeschema 7.0.6	Id: 2/2	